

8	(a)(i)1.	number of oscillations per unit time (not per second)	B1
	(a)(i)2.	$n\lambda$	A1
	(a)(ii)	$v = \text{distance} / \text{time} = n\lambda / t$ $n / t = f$, hence $v = f\lambda$	M1 A1
	(b)	1 period, $T = 3 \times 2.0 \text{ ms} = 6.0 \text{ ms}$ $f = 1 / T = 1 / (6.0 \times 10^{-3}) = 170 \text{ Hz}$	C1 A1
	(c)(i)	Sources emitting waves with a constant phase difference.	B1
	(c)(ii)	$f = v / \lambda$ $f = (3.00 \times 10^8) / (6.0 \times 10^{-2})$ $= 5.0 \times 10^9 \text{ Hz}$	C1 A1
	(c)(iii)	path difference = 3λ (integer number of wavelengths) hence no phase difference between the waves arriving at X hence constructive interference occurs $\rightarrow$ maximum intensity	M1 M1 A1
	(c)(iv)	decrease in the distance between adjacent maxima / minima	B1

	(d)(i)	<table><tr><th>order, n</th><th>angle for red maximum / °</th><th>angle for green maximum / °</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>23.6</td><td>19.0</td></tr><tr><td>2</td><td>53.0</td><td>40.5</td></tr><tr><td>3</td><td>Nil</td><td>77.2</td></tr><tr><td>4</td><td>Nil</td><td>Nil</td></tr></table> Each correct row 1 mark. (using $d \sin \theta_n = n\lambda$) (no credit for blank)	order, n	angle for red maximum / °	angle for green maximum / °	0	0	0	1	23.6	19.0	2	53.0	40.5	3	Nil	77.2	4	Nil	Nil	B4
order, n	angle for red maximum / °	angle for green maximum / °																			
0	0	0																			
1	23.6	19.0																			
2	53.0	40.5																			
3	Nil	77.2																			
4	Nil	Nil																			
	(d)(ii)	two overlapping interference patterns as red and green light have different wavelengths: <u>brightness</u> of each line/band <u>decreases</u> (since less light passes through the two slits) and the pattern of lines/bands are <u>fuzzy</u> instead of sharp/narrow	B1																		
		Any two from: - red bands wider than green bands - red bands (longer wavelength) wider apart than green bands - equally spaced fringes within each band - missing orders as the intensities are reduced by diffraction effects	B2																		
		e.g., individual double slits pattern for red (upper) and green light: 																			